RIPHAH INTERNATIONAL UNIVERSITY

DATA STRUCTURES LAB

Lab Task 12

Name: Aroosa Feroz

SAP ID: 68997

Submitted to: Ma'am Nehan Sahar

Subject: Data Structures Lab

DEPARTMENT OF COMPUTER SCIENCE

RIPHAH INTERNATIONAL UNIVERSITY, GULDBERG GREENS CAMPUS ISLAMABAD

Task 1: Identify Tree Properties

Given a tree:

```
        A
      /   \
    B       C
   / \       \
  D   E       F
```

Find:

- Root node
- Leaf nodes
- Degree of B
- Height of tree
- Level of node E

Ans:

Root Node: Root is the topmost node having no parent. Here in tree Root node is A.

Leaf Nodes: Leaf nodes are those with no children. Here D,E,F

Degree of B: Degree = number of children of a node for B it is 2 (Children: D and E)

Height of tree: Longest path from root to any leaf here it is 2

Paths:

- A → B → D = 2 edges
- A → B → E = 2 edges
- A → C → F = 2 edges

Maximum = 2 so height is 2.

Level of node E: Level starts from 0 at root

Levels:

- Level 0 → A
- Level 1 → B, C
- Level 2 → D, E, F

Here for E it is 2

Task 2: Traversals

Given:

```
        6
      /   \
    3      10
   / \
  2   5
```

Find:

1. **Inorder (LNR)**
2. **Preorder (NLR)**
3. **Postorder (LRN)**

Ans:

<u>In Order Traversal (LNR)</u>

Rule: **Left then Node then Right**

Step-by-step:

- Go to left of 6 i.e. node 3
- Go to left of 3 i.e. node 2 i.e. print 2
- Back to 3 i.e. print 3
- Go right of 3 i.e. node 5 i.e. print 5
- Back to 6 i.e. print 6
- Go right of 6 i.e. node 10 i.e. print 10

Final Answer: 2, 3, 5, 6, 10

<u>Pre Order Traversal (NLR)</u>

Rule: **Node then Left then Right**

Step-by-step:

- Start at 6 i.e. print 6
- Go left i.e. 3 i.e. print 3
- Go left i.e. 2 i.e. print 2
- Back i.e. go right i.e. 5 i.e. print 5
- Back to 6 i.e. go right i.e. 10 i.e. print 10

Final Answer: 6, 3, 2, 5, 10

<u>Post Order Traversal (LRN)</u>

Rule: **Left then Right then Node**

Step-by-step:

- Go left of 6 i.e. 3
- Go left of 3 i.e. 2 i.e. print 2
- Go right of 3 i.e.5 i.e. print 5
- Now print 3
- Go right of 6 i.e.10 i.e. print 10
- Finally print 6

Final Answer: 2, 5, 3, 10, 6

Task 3: Insert into BST:

```
1. 45, 25, 65, 10, 30
```

Show tree after each insertion.

Ans:

Rule of BST:

- If smaller go to left and if greater go right

<u>Insert 45</u>

- Tree empty so it becomes root

  Tree after insertion: 45

<u>Insert 25</u>

- 25 < 45 so go left

  Tree after insertion: 45
  /
  25

<u>Insert 65</u>

- 65 > 45 so go right

Tree after insertion:

```
    45
   /  \
25   65
```

<u>Insert 10</u>

- 10 < 45 so left
- 10 < 25 so left

Tree after insertion:

```
     45
    /  \
   25   65
  /
 10
```

<u>Insert 30</u>

- 30 < 45 so left
- 30 > 25 so right

Tree after insertion:

```
     45
    /  \
   25   65
  / \
 10  30
```

<u>Final BST:</u>

```
    45
   /  \
  25   65
  / \
 10  30
```

Thank You